



Meta adds personal boundaries

Meta to roll out a "personal boundary" system to its Horizon VR experiences, to monitor VR bullying. <https://digit.in/mar22-59>



Undiscovered tree species

Earth may have upwards of 9,200 undiscovered tree species, a third of which are in South America. <https://digit.in/mar22-60>

mRNA Technology – The Future of Health?

mRNA technology's immense potential could be a game-changer.

Chetan Bhatnagar |
feedback@digit.in

A

s time advances, humans get better at playing god. The hubris of that statement is obvious,

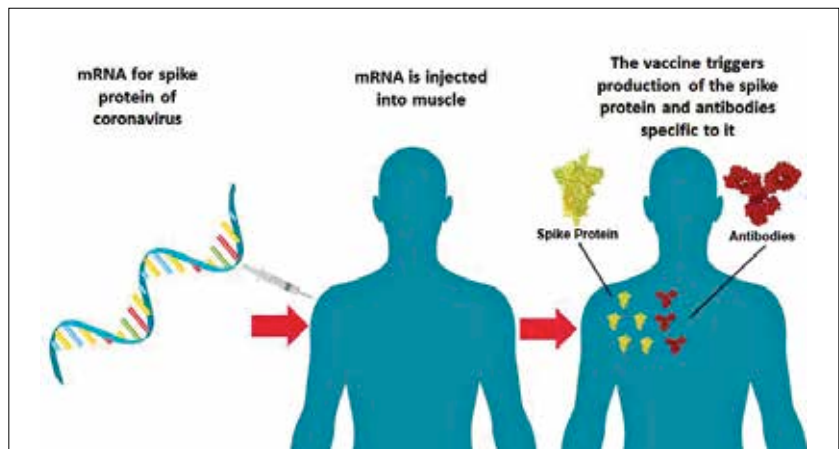
but also true. Most diseases used to spell doom for a human just over a century ago, and today we're battling viruses by extracting some of their genetic code and training our immune system to fight against it. Not only are the mRNA vaccines against COVID-19 the fastest and most efficient vaccines developed yet, but scientists aim to use the mRNA technology to fight far more than just COVID. Before we dive into the future uses of the mRNA technology, first let's take a look at why they're game changers to begin with.

BACK TO BASICS

There are a few reasons why messenger ribonucleic acid (mRNA) technology is particularly suited to vaccines. The cells in our bodies use the mRNA as a set of instructions to follow, synthesizing dynamic proteins that can treat a spectrum of diseases. The single-stranded molecule is naturally present in our cells, and carries these instructions to our cells for execution. For instance, the COVID-19 vaccines contain mRNA (it cannot "infect" us), which is then "read" by the cells upon vaccine administration. The cells that the mRNA

vaccine is administered to generate the spike protein of the COVID virus, which subsequently leads to a robust immune response (in the form of neutralizing anti-bodies). Your body then stores the memory of producing those antibodies in case it needs to deal with the spike proteins again. This is how it wakes up quickly when infected with live COVID, and fights off infection faster than a body whose immune system was not trained this way.

Perhaps, most importantly, mRNA can be engineered specifically not only to create antigens for viruses, but also to encode cytokines



The mRNA vaccine mechanism, simplified.

Image: NIH